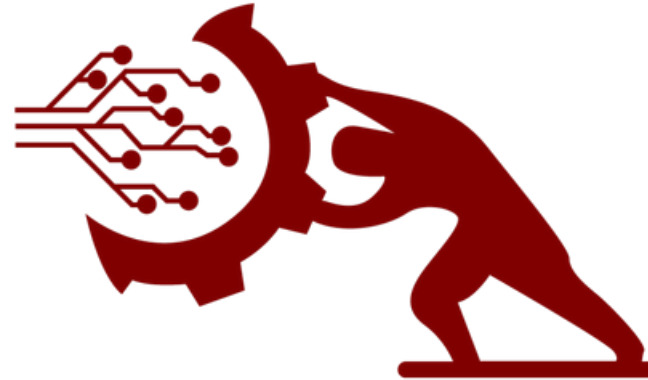


ENDEAVOUR



ENDEAVOUR ASSESSMENT 2026

Syllabus & Evaluation Scheme

<https://www.endeavourslit.in>

Endeavour Assessment 2026 — Syllabus & Evaluation Scheme

Exam: Assessment Test — *UG & ICD*

Maximum Marks: 100

Duration: 90 minutes

(note: both UG and ICD contain same marking and evaluation scheme)

Exam Structure

- **Section A:** 24 MCQs × 1 mark = 24 marks (Physics, Chemistry, Mathematics, Aptitude). Negative marking: -0.25 per wrong answer.
- **Section B:** 13 MCQs × 2 marks = 26 marks (Physics, Chemistry, Mathematics, Aptitude). Negative marking: -0.25 per wrong answer.
- **Section C:** 10 Technical MCQs × 2 marks = 20 marks. Negative marking: None
- **Section D:** 3 Descriptive questions = 30 marks. All compulsory. Questions cover project design, personal motivation, and logical reasoning.
- **Total:** 100 marks.

Evaluation Scheme

Section	Questions	Marks per Q	Section Marks	Negative Marking	Notes
Section 1	24	1	24	-0.25	Short recall MCQs
Section 2	13	2	26	-0.25	Conceptual and numerical MCQs
Section 3	10	2	20	None	Technical domain questions
Section 4	3	10	30	None	Descriptive, rubric-based
Total	50	—	100	—	—

Descriptive Rubrics

- **Q1 Project Design (10 marks):** innovation (3), technical correctness (3), clarity of plan (2), diagrams/presentation (2)
- **Q2 Motivation (10 marks):** alignment with goals (3), specific examples (3), growth plan (2), clarity of language (2)
- **Q3 Logic Puzzle (10 marks):** correct solution (4), reasoning steps (4), clarity and conciseness (2)

1. Syllabus for UG (1st + 2nd) & ICD 3rd year

1.1. Modules

- **Module 1:** Physics Fundamentals — measurement, mechanics, current electricity, magnetism & EMI, optics, modern physics, semiconductors
- **Module 2:** Chemistry Basics — moles & stoichiometry, redox reactions, electrochemistry, chemical kinetics, environmental chemistry
- **Module 3:** Mathematics Essentials — trigonometry, complex numbers, probability, differential & integral calculus, vectors, matrices & determinants
- **Module 4:** Aptitude & Reasoning — verbal reasoning, non-verbal reasoning
- **Module 5:** Technical Knowledge — sensors, Arduino, memory, communication, components
- **Module 6:** Descriptive Practice — project design, motivation writing, logic puzzle explanations

1.2. Subject Coverage

Subject	Subject Coverage	Weightage
Physics	• Measurement: dimensional analysis, significant figures, absolute/relative/percentage error • Mechanics: FBD, kinematics, work, power, energy, rotation • Current Electricity: ohm's law, series/parallel connection, battery connection, Kirchoff's laws • Magnetism&EMI: Lorentz force, current carrying conductor, Faraday's law, Lenz's law, transformers • Optics: mirror/lens, refraction, TIR • Modern Physics & Semiconductors: work function, photoelectric effect, diode, forward/reverse bias, logic gates	35%
Chemistry	• Moles & Stoichiometry: mole concept, molar mass, stoichiometry • Redox: oxidation, reduction • Electrochemistry: Galvanic cells, electrode, EMF, batteries • Chemical Kinetics: rate of reaction, catalyst, activation energy • Environmental Chemistry: lightning reactions, ozone formation, NO_x chemistry	13%

Mathematics	• Trigonometry, Complex Nos: standard identities, inverse functions, polar representation, phasors • Probability: conditional probability, Bayes theorem, independent events • Differential calculus: limits, continuity, rate of change, derivative, maxima/minima • Integral calculus: indefinite/definite integration, fundamental theorem • Vectors: Component, dot product, cross product • Matrices & Determinants: inverse, adjacent matrix, transformation	21%
Aptitudes	• Verbal Reasoning: coding-decoding, series, alphabet test, puzzle test, logical venn diagram, calendar, clock, syllogism • Non-verbal Reasoning: series completion, analogy, mirror images & water images, figure completion, figure matrix and missing figure, cube and dice	29%
Technical Knowledge	• Sensors: Ultrasonic, PIR, LDR (principle and applications) • Arduino: Pinout, PWM, board types (Uno, Nano, Mega) • Memory: EEPROM basics • Communication: IR, Bluetooth, Wi-Fi, RFID • Actuators & Modules: servo, relay, buzzer, LCD/OLED displays • Circuits: breadboarding, wiring, current limits, safety practices	10 Questions

2. Syllabus for ICD (1st & 2nd year)

2.1 Modules

- **Module 0:** Orientation & Exam Strategy — negative marking rules, time management
- **Module 1:** Physics Basics — force, work, power & energy, light, sound, electricity & magnetism
- **Module 2:** Chemistry Basics — periodic properties, acid bases & salts, mole concept
- **Module 3:** Mathematics — algebra, trigonometry, geometry, probability, series
- **Module 4:** Aptitude — verbal reasoning, non-verbal reasoning
- **Module 5:** Technical Knowledge — sensors, Arduino, EEPROM, communication
- **Module 6:** Descriptive Practice — project design, motivation writing, logic problem solving

2.2. Subject Coverage

Subject	Subject Coverage	Weightage
Physics	• Force, Work, Power & Energy: Force, turning forces, uniform circular motion, work, energy, power • Light: Snell's law, mirrors, reflection, refraction, optical phenomena • Sound: reflection of sound, echoes, and natural vibrations • Electricity & Magnetism: Current electricity, electrical power, household circuits, and electromagnetism.	35%
Chemistry	• Periodic properties: elements and trends in properties • Mole Concept: Avagadro's number, molar mass, general concepts • Carbon and its compounds: basic compounds, fullerenes	13%
Mathematics	• Algebra: quadratic polynomials, discriminant conditions, linear equations Geometry: distance formula, line division • Trigonometry: basic ratios, secant/tangent relations • Probability: valid ranges, simple problems • Series & Patterns: number series, missing terms	21%

Aptitudes	• Verbal Reasoning: coding-decoding, series, alphabet test, puzzle test, logical venn diagram, calendar, clock, syllogism • Non-verbal Reasoning: series completion, analogy, mirror images & water images, figure completion, figure matrix and missing figure, cube and dice	29%
Technical Knowledge	• Sensors: Ultrasonic, PIR, LDR (principle and applications) • Arduino: Pinout, PWM, board types (Uno, Nano, Mega) • Memory: EEPROM basics • Communication: IR, Bluetooth, Wi-Fi, RFID • Actuators & Modules: servo, relay, buzzer, LCD/OLED displays • Circuits: breadboarding, wiring, current limits, safety practices	10 Questions

#Instructor Notes

- Emphasize exam discipline (no electronic devices, no malpractice).
 - Encourage accuracy over guesswork in Sections A & B.
 - Urge full attempts in Section C since it has no penalties.
 - Provide clear practice rubrics for Section D responses.
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This syllabus and evaluation scheme outline provides UG & ICD students with a clear structure of subject coverage, section weightage, and preparation guidelines for the Endeavour Assessment 2026.
